

Case 15: Negative — DELETE Non-Existent Resource

Intermediate+

Verify the API handles DELETE requests on non-existent or invalid resources.

New concept: Combining DELETE negative testing with parametrize

API Endpoint

DELETE <https://jsonplaceholder.typicode.com/posts/99999>

What You Will Learn

- DELETE on non-existent resources
- Idempotency — what happens when you DELETE twice
- Parametrizing multiple invalid IDs for DELETE
- Combining negative DELETE with xfail

Study Plan

Step 1 — DELETE on non-existent resource

In Case 5 you deleted post 1 which exists. What happens when you DELETE /posts/99999 which doesn't exist? A real API returns 404. JSONPlaceholder returns 200 — it's fake.

Step 2 — What is idempotency?

A DELETE is idempotent — calling it twice should give the same result. First DELETE returns 200 (or 404). Second DELETE on same ID should also return 404 (already gone). This is worth testing.

Step 3 — Parametrize invalid IDs

Think: what is the changing value between each test run — the method or the ID? Parametrize the invalid IDs: 99999, -1, 0, 99999999.

Step 4 — What to assert

For non-existent IDs, a real API returns 404. JSONPlaceholder returns 200 for everything (fake). Use xfail to document this. Think: do you assert 404 or 200?

Step 5 — Think about the loop

Each parametrize run deletes one invalid ID. Inside the test, do you need a for loop? What does the DELETE response body look like — a dict or a list?

Think Before You Code

Ask yourself these questions before writing a single line:

- What is the changing value between each test run — the method or the ID?
- What type does `response.json()` return for a DELETE — a dict or a list?
- Do you need a for loop inside the test?
- What is idempotency and how would you test it with two DELETE calls?
- Should you assert 404 or 200 — and how does xfail help here?

Your Task

Write pytest tests for DELETE on non-existent and invalid resources.

Test A — Parametrized invalid IDs:

DELETE /posts/{id} for ids: 99999, -1, 0, 99999999

Mark as xfail — JSONPlaceholder returns 200 even for non-existent ids

Assert: status code is 404

Test B — Idempotency test (BONUS):

DELETE /posts/1 twice in the same test

First DELETE: assert 200

Second DELETE: assert 404 (already gone)

Mark as xfail — JSONPlaceholder won't behave this way

Before writing:

- What is the parametrize variable in Test A?
- What type does DELETE response.json() return?
- Do you need a for loop anywhere?
- For Test B, how do you make two DELETE calls in one test function?

BONUS: Print the status code and response body for each invalid ID

Hint: Test A: parametrize the id, build URL with f-string, one assert. Test B: two requests.delete() calls in sequence, two asserts — no loop needed.